

Abstract No.

Antitumor Activity of All-Trans Retinoic Acid and Curcumin-Loaded BSA Nanoparticles Against U87 Glioblastoma Cells

Ceyda Sönmez^{1,}, Aleyna Baltacıoğlu¹, Jülide Çoşkun², Gulen Melike Demirbolat³, Ozgul Gok⁴, and Aysel Özpınar^{1,5}*

¹Department of Biochemistry and Molecular Biology, Graduate School of Health Sciences, Acibadem Mehmet Ali Aydınlar University, Istanbul, Türkiye

²Acibadem Labmed Clinical Laboratories, Istanbul 34752, Türkiye

³Department of Pharmaceutical Technology, Faculty of Pharmacy, Acibadem Mehmet Ali Aydınlar University, Istanbul 34752

⁴Department of Biomedical Engineering, Faculty of Engineering and Natural Sciences, Acibadem Mehmet Ali Aydınlar University, Istanbul 34752, Türkiye

⁵Department of Medical Biochemistry, School of Medicine, Acibadem Mehmet Ali Aydınlar University, Istanbul, Türkiye

*Presenting author

Introduction: Glioblastoma (GBM) is the most aggressive primary brain tumor, characterized by intrinsic drug resistance, invasive growth, and the presence of the blood–brain barrier (BBB), which together render standard therapies largely ineffective. All-trans retinoic acid (ATRA) and curcumin (CURC) are potent antitumor agents that target proliferation, invasion, inflammation, and oxidative stress through distinct mechanisms; however, their clinical utility is limited by poor bioavailability and rapid clearance. This study aimed to develop a bovine serum albumin (BSA) nanoparticle (NP) platform for the co-delivery of ATRA and CURC and to evaluate its antitumor activity in U87-MG GBM cells.

Methods: ATRA/CURC-loaded BSA-NPs were synthesized and characterized for particle size, morphology, encapsulation efficiency, and drug release behavior. Cytotoxic and antimigratory effects were assessed in U87-MG cells using real-time xCELLigence proliferation and migration assays, while apoptosis was evaluated by Annexin V/PI staining and flow cytometry at 24, 48, and 72 h.

Results: The combination NPs exhibited a mean particle size of 154.7 nm and an encapsulation efficiency 99%, both favorable for BBB permeability. Compared with single-drug formulations, ATRA-CURC-BSA-NPs produced the most pronounced suppression of cell proliferation and the strongest, most sustained inhibition of migration. Apoptosis assays revealed a time-dependent shift from early to late apoptosis, with the combination treatment yielding the highest apoptotic fraction at 72 h, substantially exceeding single-agent groups.

Conclusion: Co-delivery of ATRA and CURC via BSA nanoparticles enhances antiproliferative, antimigratory, and pro-apoptotic effects in U87-MG GBM cells compared with single-drug delivery. These findings highlight albumin-based co-delivery of ATRA and CURC as a promising strategy to overcome current therapeutic limitations in glioblastoma and provide a rationale for future translational and in vivo investigations.

Keywords: glioblastoma; U87-MG; all-trans retinoic acid; curcumin; BSA nanoparticles; drug delivery; apoptosis; cell migration



Supported by SciSerTec

Web: <http://www.scisertec.de>